SEQUENCE 3

Straight lines in the plane

1–16 October 2026

3.1 Direction vector and Cartesian equation

Key point · Definition. A direction vector of a line is a **nonzero** vector having the direction of that line. If $A \neq B$, $\overrightarrow{AB}$ directs (AB). Every line has an equation $ax + by + c = 0$ with $(a, b) \neq (0, 0)$. A direction vector is $\begin{pmatrix} -b \\ a \end{pmatrix}$.

Example 1. For $A(-2, 4)$ and $B(6, 2)$, calculate $\overrightarrow{AB}$ then check that $\vec{u}(4, -1)$ directs (AB).

Key point · Establishing an equation with the determinant. The line through $A(x_A, y_A)$ with direction vector $\vec{u}(\alpha, \beta) \neq \vec{0}$ has equation

$$\beta(x - x_A) - \alpha(y - y_A) = 0.$$

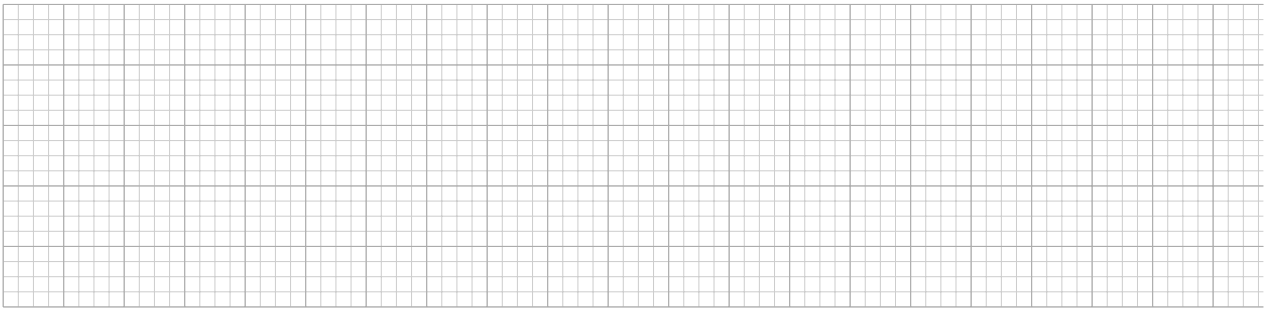
Proof to write.

Watch out. The zero vector directs no line. The sign of the determinant depends on the order of the vectors, but whether it vanishes does not.

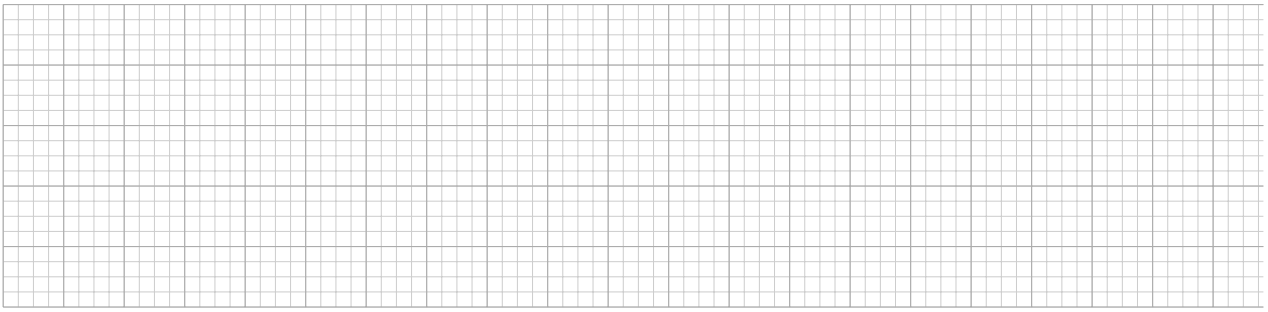
3.2 Determining the equation of a line

Method. From a point and a direction vector: choose a, b then replace x, y by the coordinates of the point to find c. From two distinct points: first calculate a direction vector.

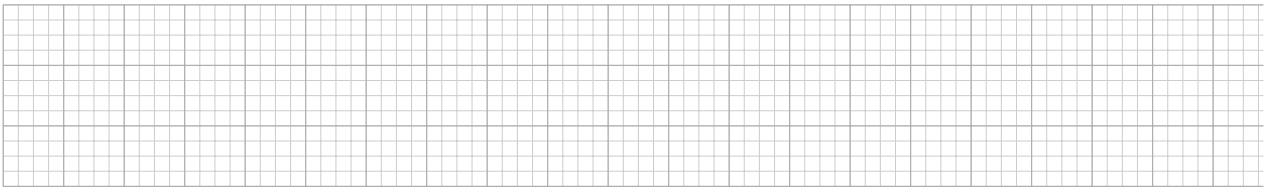
Example 2. Determine a Cartesian equation of the line through $A(2, -1)$ with direction vector $\vec{u}(3, 2)$.



Example 3. Determine a Cartesian equation of (AB) for $A(3, 3)$ and $B(2, -2)$.



Example 4. Does the point $P(1, 2)$ belong to $d : 2x - 3y + 4 = 0$? Does the point $Q(2, 1)$ belong to it?

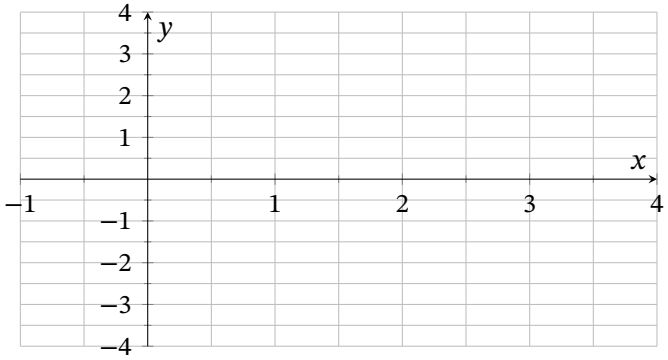


3.3 Slope-intercept equation, slope and affine function

Key point · Definitions. A non-vertical line has a unique slope-intercept equation $y = mx + p$. m is the slope, p the y-intercept. It represents the affine function $x \mapsto mx + p$; $(1, m)$ is a direction vector. A vertical line has an equation $x = k$; it is not the graph of a function of x. A horizontal line has an equation $y = k$.

Key point · Calculating a slope. If $x_A \neq x_B$, the slope of (AB) is $m = \dfrac{y_B - y_A}{x_B - x_A}$. To find p, we then use $y_A = mx_A + p$.

Example 5. Determine the slope-intercept equation of $2x + 3y - 1 = 0$, then draw $d : y = -2x + 3$.



Example 6. Find the affine function f such that $f(0) = 3$ and $f(-1) = 5$, then the affine function g whose graph passes through $C(-1, 2)$ and $E(5, -2)$.

3.4 Relative positions of two lines

Key point · Properties. Two lines are either intersecting, strictly parallel or coincident. For $y = mx + p$ and $y = m'x + p'$:

- if $m \neq m'$, they intersect;
- if $m = m'$ and $p \neq p'$, they are strictly parallel;
- if $m = m'$ and $p = p'$, they coincide.

For Cartesian equations, we compare the directions with the determinant. If the directions are collinear, we then test the membership of a point to distinguish coincident lines from strictly parallel ones.

Example 7. Determine an equation of d through $A(2, 3)$ and parallel to $d' : x - 3y - 3 = 0$.

Example 8. Study the relative positions:

1. $2x - y + 1 = 0$ and $4x - 2y + 2 = 0$;
2. $x = 2$ and $x = -1$;
3. $y = 2x - 3$ and $y = -x + 3$.

3.5 Intersection: solving by substitution

Key point · Definition. Solving a system of two equations in x, y means finding all the pairs (x, y) satisfying both equations simultaneously. For two equations of lines: intersecting lines give one solution pair; strictly parallel lines give no solution; coincident lines give infinitely many solutions.

Method · Substitution. Isolate one unknown in one equation, substitute its expression into the other, solve, then find the second unknown. Check the pair in both original equations.

Example 9. Solve $\begin{cases} -2x + y + 3 = 0, \\ 3x - 5y - 2 = 0. \end{cases}$

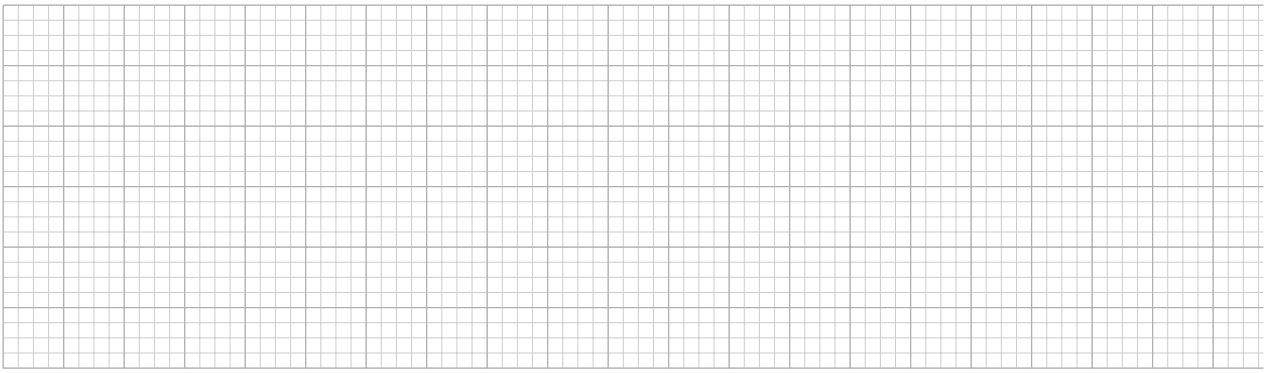
Example 10. Solve by substitution:

(a) $\begin{cases} -2x + y + 3 = 0 \\ x - 4y + 5 = 0 \end{cases}$ (b) $\begin{cases} 2x + 3y - 10 = 0 \\ 4x + 5y - 16 = 0. \end{cases}$

3.6 Linear combination and Python

Method · Linear combination. Multiply the equations by nonzero numbers to obtain opposite coefficients for one unknown; add them to eliminate that unknown. Keep one equation of the system to calculate the other unknown.

Example 11. Solve $\begin{cases} 2x+3y+1=0 \\ 3x-4y-2=0. \end{cases}$ Then redo the two systems of the previous example by linear combination.



```
def determinant(u, v):
    return u[0]*v[1] - u[1]*v[0]

def alignes(A, B, C):
    AB = (B[0]-A[0], B[1]-A[1])
    AC = (C[0]-A[0], C[1]-A[1])
    return determinant(AB, AC) == 0
```

Example 12. Predict `alignes((0,0), (2,1), (6,3))` then `alignes((0,0), (2,1), (6,4))`.

